

# FAULT CODE 238 (Source Address 00) - Sensor Supply 3 Circuit - Voltage Below Normal or Shorted to Low Source

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## Warnings and Cautions

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### CAUTION

To reduce the possibility of damaging a new ECM, all other active fault codes must be investigated prior to replacing the ECM.

### CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead, Part Number 3822917 - female Deutsch™/AMP™/Metri-Pack™ test lead, Part Number 3164596 - male Framatome™ test lead, and Part Number 3164597 - female Framatome™ test lead.

## Troubleshooting Summary

| STEPS   |                                                                                                                             | SPECIFICATIONS                                                                        |
|---------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| STEP 1. | Check the fault codes.                                                                                                      |                                                                                       |
|         | <b>STEP 1A.</b> Check for an active fault code.                                                                             | Fault Code 238 active or inactive with more than 1 count in the last 25 engine hours? |
| STEP 2. | Check the sensors and circuits connected to the sensor supply 3 and return.                                                 |                                                                                       |
|         | <b>STEP 2A.</b> Inspect the engine crankshaft speed/position sensor and circuit connected to the sensor supply 3 and return | Dirty or damaged pins?                                                                |
|         | <b>STEP 2B.</b> Check the circuit response.                                                                                 | Fault Code 238 active?                                                                |

| STEPS          |                                                                           | SPECIFICATIONS                                                                                          |
|----------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>STEP 3.</b> | Check the ECM.                                                            |                                                                                                         |
|                | <b>STEP 3A.</b> Inspect the ECM and engine wiring harness connector pins. | Dirty or damaged pins?                                                                                  |
|                | <b>STEP 3B.</b> Check the engine control module (ECM) response.           | Fault Code 238 active?                                                                                  |
| <b>STEP 4.</b> | Clear the fault codes.                                                    |                                                                                                         |
|                | <b>STEP 4A.</b> Check if an ECM calibration update is available.          | If a calibration update for this fault code is available, does the ECM contain that revision or higher? |
|                | <b>STEP 4B.</b> Disable the fault code.                                   | Fault Code inactive?                                                                                    |

## STEP 1. Check for an active fault code.

### STEP 1A. Check for an active fault code.

#### Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

| Action                                                                                                                                           | Specification/Repair                                                                                    | Next Step              |
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| Check for an active fault code. <ul style="list-style-type: none"> <li>• Use INSITE™ electronic service tool to read the fault codes.</li> </ul> | Fault Code 238 active or inactive with more than 1 count in the last 25 engine hours?<br><br><b>YES</b> | <b>2A</b>              |
|                                                                                                                                                  | Fault Code 238 active or inactive with more than 1 count in the last 25 engine hours?<br><br><b>NO</b>  | <b>Repair Complete</b> |

**STEP 2. Check the sensors and circuits connected to the sensor supply 3 and return.**

**STEP 2A. Inspect the engine crankshaft speed/position sensor connector and circuit connected to the sensor supply 3 and return.**

- Conditions :**
- Turn keyswitch OFF.
  - Disconnect the engine crankshaft speed/position sensor connector from the engine wiring harness connector.

| Action | Specification/Repair | Next Step |
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| <p>Inspect the engine wiring harness and engine crankshaft speed/position sensor connector pins for the following:</p> <ul style="list-style-type: none"> <li>• Loose connector</li> <li>• corroded pins</li> <li>• Bent or broken pins</li> <li>• Pushed back or expanded pins</li> <li>• Moisture in or on the connector</li> <li>• Missing or damaged connector seals</li> <li>• Dirt or debris in or on the connector pins</li> <li>• Wire insulation damage</li> <li>• Connector shell broken</li> <li>• Damaged connector locking tab.</li> </ul> <p>Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19.<br/>(/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)</p> | <p>Dirty or damaged pins?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>A damaged connection had been detected in the engine crankshaft speed/position sensor or wiring harness connector.</p> <p>Clean the connector and pins.</p> <ul style="list-style-type: none"> <li>• Repair or replace the damaged section of the engine wiring harness. Refer to Procedure 019-043 in the Associated Procedures Table.</li> <li>• Repair or replace the damaged engine crankshaft speed/position sensor. Refer to Procedure 019-042 in the Associated Procedure Table.</li> </ul> | <p><b>4A</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Dirty or damaged pins?</p> <p><b>NO</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p><b>2B</b></p> |

## STEP 2B. Check the circuit response.

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>• Turn keyswitch OFF.</li> <li>• Disconnect the engine crankshaft speed/position sensor connector from the engine wiring harness connector.</li> <li>• Turn keyswitch ON.</li> <li>• Connect INSITE™ electronic service tool.</li> </ul> |                      |           |
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| Action                                                                                                                                                                                                                                                                                                     | Specification/Repair | Next Step |

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| <p>Check for the appropriate ECM response after 30 seconds.</p> <ul style="list-style-type: none"> <li>• Use INSITE™ electronic service tool to read the fault codes.</li> </ul> | <p>Fault Code 238 active?</p> <p><b>YES</b></p>                                                                                                                                                                                                      | <b>3A</b> |
|                                                                                                                                                                                  | <p>Fault Code 238 active?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <ul style="list-style-type: none"> <li>• Replace the damaged engine crankshaft speed/position sensor. Refer to Procedure 019-042 in the Associated Procedure Table.</li> </ul> | <b>4A</b> |

## STEP 3. Check the ECM.

### STEP 3A. Inspect the ECM and engine wiring harness connector pins.

**Conditions :**

- Turn keyswitch OFF.
- Disconnect the engine wiring harness connector from the ECM 60-pin connector.

| Action | Specification/Repair | Next Step |
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| <p>Inspect the engine wiring harness and ECM connector pin for the following:</p> <ul style="list-style-type: none"> <li>• Loose connector</li> <li>• Corroded pins</li> <li>• Bent or broken pins</li> <li>• Pushed back or expanded pins</li> <li>• Moisture in or on the connector</li> <li>• Missing or damaged connector seals</li> <li>• Dirt or debris in or on the connector pins</li> <li>• Connector shell broken</li> <li>• Wire insulation damage</li> <li>• Damaged connector locking tab.</li> </ul> <p>Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19.<br/>(/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)</p> | <p>Dirty or damaged pins?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>A damaged connection has been detected in the ECM connector or the engine wiring harness.</p> <p>Clean the connector and pins.</p> <p>Refer to the circuit diagram or wiring diagram for all wiring harness interconnections.</p> <ul style="list-style-type: none"> <li>• Repair or replace only the components and/or damaged section the engine wiring harness. Refer to Procedure 019-043 in the Associated Procedures Table.</li> </ul> <p>Refer to Procedure 019-031 in Section 19.<br/>(/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)</p> <p>Dirty or damaged pins?</p> <p><b>NO</b></p> | <p><b>4A</b></p> <p><b>3B</b></p> |
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### STEP 3B. Check the ECM response.

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>• Turn keyswitch OFF.</li> <li>• Disconnect the engine wiring harness connector from the ECM 60-pins connector.</li> <li>• Turn keyswitch ON.</li> <li>• Connect INSITE™ electronic service tool.</li> </ul> |                      |           |
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| Action                                                                                                                                                                                                                                                                         | Specification/Repair | Next Step |

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| <p>Inspect the engine wiring harness and ECM connector pin for the following:</p> <ul style="list-style-type: none"> <li>• Loose connector</li> <li>• Corroded pins</li> <li>• Bent or broken pins</li> <li>• Pushed back or expanded pins</li> <li>• Moisture in or on the connector</li> <li>• Missing or damaged connector seals</li> <li>• Dirt or debris in or on the connector pins</li> <li>• Connector shell broken</li> <li>• Wire insulation damage</li> <li>• Damaged connector locking tab.</li> </ul> <p>Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19.<br/>(/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)</p> | <p>Fault Code 238 active?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the ECM. Refer to Procedure 019-031 in Section 19.<br/>(/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)</p>                                                                                                                                                                                                                                                                                                                            | <p><b>4A</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Fault Code 238 active?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Troubleshoot each section of the wiring harnesses connected in series to determine which contains the short circuit to ground or pin-to-pin short circuit.</p> <p>Refer to the circuit diagram or wiring diagram for all wiring harness interconnections.</p> <ul style="list-style-type: none"> <li>• Repair or replace the damaged section of the engine wiring harness. Refer to Procedure 019-043 in the Associated Procedures Table.</li> </ul> | <p><b>4A</b></p> |

## STEP 4. Check ECM calibration and clear fault codes.

### STEP 4A. Check if an ECM calibration update is available.

**Conditions :**

- Connect all components.
- Connect INSITE™ electronic service tool.

| Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Specification/Repair                                                                                                                                                                                                                                                                  | Next Step |
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| Compare the ECM code and revision number in the ECM to the calibration revisions listed in the ECM Calibration Revision History for applicable changes related to this fault code. <ul style="list-style-type: none"><li>• Use INSITE™ electronic service tool to find the present ECM code and revision number in the ECM. The ECM code and revision number are found in the Calibration Information section of System ID and Dataplate in Features and Parameters.</li></ul> | If a calibration update for this fault code is available, does the ECM contain that revision or higher?<br><br><b>YES</b>                                                                                                                                                             | <b>4B</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | If a calibration update for this fault code is available, does the ECM contain that revision or higher?<br><br><b>NO</b><br><br><b>Repair :</b><br><br>If necessary, calibrate the ECM. Refer to Procedure 019-032 in Section 19 (/qs3/pubsys2/xml/en/procedures/99/99-019-032.html). | <b>4B</b> |

**STEP 4B. Disable the fault code.****Conditions :**

- Connect all components.
- Connect INSITE™ electronic service tool.

| Action | Specification/Repair | Next Step |
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| Disable and clear the fault code.<br><br><ul style="list-style-type: none"> <li>Operate the engine within the "Conditions for Clearing the Fault Code" found in the Overview section of this troubleshooting procedure.</li> </ul> | Fault Code inactive?<br><br><b>YES</b>                                                                                                                                                          | <b>Repair complete</b>                 |
|                                                                                                                                                                                                                                    | Fault Code inactive?<br><br><b>NO</b><br><br><b>Repair :</b><br><br>Verify that all steps have been completed. If all steps have been completed, then follow your technical escalation process. | <b>Escalate or call for assistance</b> |

| Associated Procedures     |                                                                                     |                    |                 |
|---------------------------|-------------------------------------------------------------------------------------|--------------------|-----------------|
| Procedure Title           | Procedure Number                                                                    | Service Model Name | Bulletin Number |
| Engine Speed Sensor (ESS) | Refer to Procedure 019-042<br>(/qs3/pubsys2/xml/en/procedures/122/122-019-042.html) | QSK50 CM2150 MCRS  | 4022102         |
| Engine Wiring Harness     | Refer to Procedure 019-043<br>(/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) | QSK50 CM2150 MCRS  | 4022102         |